

# Ziyi Sun

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College of Engineering, Ocean University of China, Qingdao 266100, China

## EDUCATION

### • Ocean University of China (985 / Double First-Class)

Sep. 2023 – Jun. 2027 (Expected)

B.Eng. in Mechanical Design, Manufacturing and Automation

Qingdao, China

◦ GPA: 3.7/4.0 Rank: 1/56 Average: 89.2

◦ Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

## RESEARCH EXPERIENCE

### • Roboparty Lab

Jul. 2026 – Oct. 2026

Frontier Research and Open Source Intern (Mentor: Jagger)

Beijing, China

◦ Contribute to frontier research and open-source development for embodied intelligence and robot learning systems; related work: [S.1].

## PROJECTS

### • Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection

Dec. 2025 – Dec. 2026

Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing

OUC SRDP

◦ Designed a biomimetic robotic fish with adjustable body length (1.2–1.5 m) and fused visual tracking with water-quality sensing for aquaculture monitoring.

### • Cross-Domain Wire-Driven Biomimetic Eel Robot

Dec. 2024 – Nov. 2025

Role: Project Leader | Tools: vision, Gaussian Splatting, damage detection

Provincial | [C.1], [C.2]

◦ Built an eel-like robot for offshore wind dynamic-cable inspection with visual tracking, 3D reconstruction, and damage detection; led to IEEE RCAR / ICCCR publications.

### • Streaming Evaluation Framework for Deep-Sea Mining

2025 – 2026

Role: Lead Contributor | Tools: CAD modeling, multi-AUV simulation

Research | [S.2]

◦ Built a unified simulation and benchmarking pipeline comparing centralized vs. distributed multi-AUV nodule collection for underwater robot learning.

### • High-Stability Duck-Type Wave Energy Converter with Pendulum PTO

Dec. 2024 – Nov. 2025

Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks

National | [P.1]

◦ Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and adaptive PTO design (first student inventor on related patent).

## PUBLICATIONS & PATENTS

C=CONFERENCE, S=IN SUBMISSION, P=PATENT, SW=SOFTWARE

[C.1] Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng\*.

**A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.**

IEEE RCAR, 2026. **Best Paper Finalist.**

[C.2] Ziyi Sun et al.

**Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.**

IEEE ICCCR, 2025.

[S.1] Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.

**CHEROE: Every Humanoid Skill as a Traject.**

Under review at AAAI 2027 (CCF-A).

[S.2] Ziyi Sun, Changhong He.

**A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.**

Under review at *Journal of Marine Science and Engineering (JMSE)*.

[P.1] Ziyi Sun et al.

**A High-Stability PTO System for Duck-Type Wave Energy Converters.**

Invention Patent (First Student Inventor).

[SW.1] Ziyi Sun.

**Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.**

Software Copyright (No. 2024SR0140235).

SKILLS

- **Robot Platform:**Unitree G1
- **Programming & ML:**Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:**SolidWorks, AutoCAD
- **Research Focus:**Embodied AI, robot learning, sim-to-real transfer, multimodal perception, underwater robotics
- **Languages:**Chinese (Native); English (CET-4: 591, CET-6: 482)

HONORS AND AWARDS

- **Shandong Provincial Government Scholarship**2025
- **First-Class Comprehensive Scholarship, Ocean University of China**2024 & 2025
- **Outstanding Student / Outstanding Student Cadre, Ocean University of China**2024 & 2025
- **National First Prize, National 3D Digital Innovation Design Competition**2025 & 2026
- **National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)**2025
- **Honorable Mention, Mathematical Contest in Modeling (MCM)**2025
- **National Third Prize, 18th National College Student Energy Conservation Competition**2024
- **National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest**2024
- **National Second Prize, 14th Marine Vehicle Design and Production Competition**2025
- **National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition**2026
- **Science and Technology Innovation Pioneer Team, College of Engineering, OUC**2024

LEADERSHIP EXPERIENCE

- **Class Representative / Class Monitor, Ocean University of China**2023 – Present
- **Team Leader, College of Engineering Debate Team, OUC**2023 – Present
- **Director of Event Organization, Student Association for the Study of Socialism with Chinese Characteristics, OUC**2023 – Present

VOLUNTEER EXPERIENCE

- **Volunteer, Qingdao Marathon**2023
- **Member, National College Student Volunteer Outreach Team (Zunyi Conference)**2024
- **Outstanding Individual in Summer Social Practice, Ocean University of China**2024